Formalization of Weingarten Calculus

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Chapter 1

Schur Weyl Duality

1.1 Symmetric subspaces

Lemma 1.1.1. *Test lemma*

Theorem 1.1.2. *Test theorem*

Suppose $\mathcal{X} = \mathbb{C}^{[d]}$ and $\mathcal{Y} = \mathbb{C}^{[d]}$ where $d \in \mathbb{N}$. Let the set of linear operators be $\mathcal{L}(\mathcal{X}, \mathcal{Y})$.

Definition 1.1.3 (symmetric subspace of linear operators)**.** The symmetric subspace of linear operators $\mathcal{L}(\mathcal{X})^{\circledS k}$ is the set of all operators $X \in \mathcal{L}(\mathcal{X}^{\otimes k})$ that remain unchanged (invariant) under the permutation W_π. That is,

$$\mathcal{L}(\mathcal{X})^{\circledS k} = \{X \in \mathcal{L}(\mathcal{X}^{\otimes k}) : X = W_\pi X W_\pi^* \text{ for all } \pi \in S_k\}. \tag{1.1}$$

Remark 1.1.4. Let S_k denote the symmetric group. Then given $\pi \in S_k$, the permutation matrix (operator) $V_d(\pi)$ is the unitary matrix that satisfies

$$W_\pi |\psi_1\rangle \otimes \ldots |\psi_k = |\psi_{\pi^{-1}(1)}\rangle \otimes \cdots \otimes |\psi_{\pi^{-1}(k)}\rangle \tag{1.2}$$

for all $|\psi_1\rangle, \ldots, |\psi_k\rangle \in \mathbb{C}^d$.